Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Complex roots of a quadratic**

Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.

Its discriminant is $\Delta = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0$. A negative discriminant means the quadratic formula takes the square root of a negative number, giving a complex conjugate pair (Theorem 1.1) instead of real roots.

2 **[Foundational]** • **Target: Complex roots of a quadratic**

Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.

$(-3 + 2i)^2 = 9 - 12i + 4i^2 = 9 - 12i - 4 = 5 - 12i$. And $6(-3 + 2i) = -18 + 12i$. Adding: $(5 - 12i) + (-18 + 12i) + 13 = 0$. Yes, it checks out.

3 **[Foundational]** • **Target: Modulus**

Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).

$|z| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$.

4 **[Foundational]** • **Target: Modulus**

Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.

$|a + bi| = \sqrt{a^2 + b^2}$ and $|a - bi| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2}$ — squaring b or $-b$ gives the same result, so the two moduli are identical.

5 **[Foundational]** • **Target: Complex roots of a quadratic**

Evaluate the expression $\frac{(3 + 2i)^2}{i}$ and write the result in standard form $a + bi$.

1. Expand the numerator: $(3 + 2i)^2 = 3^2 + 2(3)(2i) + (2i)^2 = 9 + 12i - 4 = 5 + 12i$. 2. Divide by i by multiplying numerator and denominator by $-i$ (or using $1/i = -i$): $\frac{5 + 12i}{i} = (5 + 12i)(-i) = -5i - 12i^2 = -5i + 12 = 12 - 5i$. In standard form $a + bi$, the answer is $12 - 5i$.

6 **[Foundational]** • **Target: Modulus**

[Complex Numbers · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

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7 [Foundational] • Target: Arithmetic with complex numbers

Solve the quadratic equation $2x^2 - 6x + 9 = 0$ over the complex numbers, and state the sum and product of the roots.

Step 1: Compute the discriminant: $\Delta = b^2 - 4ac = (-6)^2 - 4(2)(9) = 36 - 72 = -36$. Step 2: Apply the quadratic formula: $x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{6 \pm \sqrt{-36}}{2(2)} = \frac{6 \pm 6i}{4} = \mathbf{\frac{3}{2} \pm \frac{3}{2}i}$. Step 3: Verify using Vieta's formulas: Sum of roots: $\left(\frac{3}{2} + \frac{3}{2}i\right) + \left(\frac{3}{2} - \frac{3}{2}i\right) = 3 = -\frac{b}{a} = -\frac{-6}{2} = 3$ ✓. Product of roots.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Complex roots of a quadratic

An arithmetic sequence has third term $a_3 = 11$ and eighth term $a_8 = 31$. Find the common difference d, the first term a_1, and the sum of the first 20 terms S_{20}.

1. Express the terms: $a_3 = a_1 + 2d = 11$ and $a_8 = a_1 + 7d = 31$. 2. Subtract the equations: $(a_1 + 7d) - (a_1 + 2d) = 5d = 20 \Rightarrow d = 4$. 3. Find a_1: $a_1 + 2(4) = 11 \Rightarrow a_1 = 3$. 4. Compute S_{20}: $S_{20} = \frac{20}{2}[2(3) + (20-1)(4)] = 10[6 + 19(4)] = 10[6 + 76] = 10(82) = 820$.

9 [Intermediate] • Target: Modulus

Let $z = \frac{5+2i}{2-3i} + i^{43}$. (a) Express z in standard form $a + bi$. (b) Compute the modulus $|z|$.

(a) Simplify the quotient and power of i:First evaluate $\frac{5+2i}{2-3i}$: multiply numerator and denominator by the conjugate $2 + 3i$: $\frac{(5+2i)(2+3i)}{(2-3i)(2+3i)} = \frac{10 + 15i + 4i + 6i^2}{4 - 9i^2} = \frac{10 - 6 + 19i}{4 + 9} = \frac{4 + 19i}{13}$. Now evaluate i^{43}: $43 = 4(10) + 3$, remainder 3, so $i^{43} = i^3 = -i = -\frac{13i}{13}$.Combine terms: $z = \frac{4 + 19i}{13} - i = \frac{4 + 19i - 13i}{13} = \frac{4 + 6i}{13} = \mathbf{\frac{4}{13} + \frac{6}{13}i}$. (b) Modulus $|z|$.

10 [Intermediate] • Target: Arithmetic with complex numbers

Express $z = \frac{4+7i}{1-2i}$ in standard form $a + bi$, and compute its modulus squared $|z|^2$.

Step 1: Multiply numerator and denominator by the conjugate $1 + 2i$: $z = \frac{(4+7i)(1+2i)}{(1-2i)(1+2i)} = \frac{4(1) - 7(2) + [4(2) + 7(1)]i}{1^2 + (-2)^2}$ Step 2: Simplify numerator and denominator: $z = \frac{4 - 14 + (8+7)i}{1+4} = \frac{-10 + 15i}{5} = \mathbf{-2 + 3i}$. Step 3: Compute modulus squared: $|z|^2 = (-2)^2 + 3^2 = 4 + 9 = \mathbf{13}$. (Verification via quotient property: $|z|^2 = \frac{|4+7i|^2}{|1-2i|^2} = \frac{4^2 + 7^2}{1^2 + (-2)^2} = \frac{16 + 49}{1 + 4} = \frac{65}{5} = 13$ ✓).

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11 [Intermediate] • Target: Complex roots of a quadratic

The sum of the first n terms of an arithmetic sequence is given by $S_n = 2n^2 + 3n$. Find the 10th term a_{10}.

Step 1: Use relationship between S_n and a_n: $a_n = S_n - S_{n-1}$ Step 2: Method 1 (Direct evaluation): $S_{10} = 2(10^2) + 3(10) = 2(100) + 30 = 230$ $S_9 = 2(9^2) + 3(9) = 2(81) + 27 = 162 + 27 = 189$ $a_{10} = S_{10} - S_9 = 230 - 189 = \mathbf{41}$. Method 2 (General nth term formula): $a_n = (2n^2 + 3n) - [2(n-1)^2 + 3(n-1)] = (2n^2 + 3n) - (2n^2 - 4n + 2 + 3n - 3) = 4n + 1$ For $n = 10$: $a_{10} = 4(10) + 1 = \mathbf{41}$.

12 [Intermediate] • Target: Modulus

Given $f(x) = \frac{2x+5}{x-3}$ for $xeq3$, find the inverse function $f^{-1}(x)$ and state its domain.

1. Set $y = \frac{2x+5}{x-3}$ and swap x and y: $x = \frac{2y+5}{y-3}$. 2. Multiply by $y - 3$: $x(y-3) = 2y + 5 \Rightarrow xy - 3x = 2y + 5$. 3. Collect y terms: $xy - 2y = 3x + 5 \Rightarrow y(x-2) = 3x + 5$. 4. Divide by $x - 2$: $f^{-1}(x) = \frac{3x+5}{x-2}$. The domain of $f^{-1}(x)$ excludes $x = 2$: $(-\infty, 2) \cup (2, \infty)$. (This matches the range of f, which has horizontal asymptote $y = 2$).

13 [Intermediate] • Target: Arithmetic with complex numbers

A committee of 4 people is to be selected from a group of 6 men and 5 women. What is the probability that the committee consists of exactly 2 men and 2 women?

1. Find the total number of ways to choose 4 people from 11: $\binom{11}{4} = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} = 330$. 2. Find the number of ways to choose 2 men from 6 and 2 women from 5: $\binom{6}{2} \times \binom{5}{2} = 15 \times 10 = 150$. 3. Compute the probability: $P = \frac{150}{330} = \frac{15}{33} = \frac{5}{11} \approx 0.4545$.

14 [Intermediate] • Target: Complex roots of a quadratic

Let $A = (4, 3; 2, 2)$. Find A^{-1} and compute the vector $\mathbf{x} = A^{-1}(10; 6)$.

1. Compute $\det(A) = (4)(2) - (3)(2) = 8 - 6 = 2$. 2. Form A^{-1}: $A^{-1} = \frac{1}{2}(2, -3; -2, 4) = (1, -1.5; -1, 2)$. 3. Multiply by $(10; 6)$: $\mathbf{x} = \frac{1}{2}(2(10) + (-3)(6); -2(10) + 4(6)) = \frac{1}{2}(20 - 18; -20 + 24) = \frac{1}{2}(2; 4) = (1; 2)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Modulus • Cross-Lesson: 6.01 Probability

A student answers 35 questions in 42 minutes. To finish all 50 questions on the exam, how many seconds per question must the student maintain for the remaining questions?

The exam has $60 \text{ minutes} = 3600 \text{ seconds}$ total. 42 minutes is $42 \times 60 = 2520 \text{ seconds}$. The remaining time is $3600 - 2520 = 1080 \text{ seconds}$. The remaining number of questions is $50 - 35 = 15$. The allowed pacing is $\frac{1080 \text{ seconds}}{15 \text{ questions}} = 72 \text{ seconds per question}$.

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16 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 2.01 Sequences & Series

Find the equation of the tangent line to the parabola $f(x) = 2x^2 - 3x + 1$ at the point where $x = 2$.

1. Find the point on the curve: $f(2) = 2(2)^2 - 3(2) + 1 = 8 - 6 + 1 = 3$. The point is $(2, 3)$. 2. Compute the derivative: $f'(x) = 4x - 3$. 3. Evaluate the slope at $x = 2$: $m = f'(2) = 4(2) - 3 = 5$. 4. Write tangent line using point-slope form: $y - 3 = 5(x - 2) \Rightarrow y - 3 = 5x - 10 \Rightarrow y = 5x - 7$.

17 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 3.01 Functions & Graphs

[Test format & strategy · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

18 [Challenging] • Target: Modulus • Cross-Lesson: 4.01 Matrices & Vectors

[Functions & Graphs · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

19 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 5.01 Introductory Calculus

[Introductory Calculus · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

20 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 6.01 Probability

[Matrices & Vectors · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Arithmetic with complex numbers

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.

a. **6** ✓

b. 8

c. 10

d. 12

The quadratic equation is $x^2 - 4x + 5 = 0$. By Vieta's formulas, the sum of the roots is $z_1 + z_2 = -(-4)/1 = 4$ and the product of the roots is $z_1 z_2 = 5/1 = 5$. We are interested in $w = z_1^2 + z_2^2$. We can rewrite this as $w = (z_1 + z_2)^2 - 2z_1 z_2$. Substituting the values from Vieta's formulas: $w = (4)^2 - 2(5)$ $w = 16 - 10$ $w = 6$. So, $w = 6$. When expressed in the form $a + bi$, we have $a = 6$ and $b = 0$. The question asks for $a - b$. $a - b = 6 - 0 = 6$.

22 [Foundational] • Target: Complex roots of a quadratic

Find the product of the roots of $x^2 - 8x + 65 = 0$.

a. **65** ✓

b. 67

c. -8

d. -65

By Vieta's formulas, the product of the roots of $x^2 + Bx + C = 0$ is C; here that's 65 (the imaginary parts of the conjugate pair cancel in the sum, and multiply to a real number in the product).

23 [Foundational] • Target: Modulus

Find the product of the two roots of $x^2 - 2x + 10 = 0$.

a. 10 ✓

b. 1

c. -9

d. 2

By Vieta's formulas the product of the roots is $c/a = 10/1 = 10$ (also checkable directly: $(1 + 3i)(1 - 3i) = 1 + 9 = 10$).

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24 [Foundational] • Target: Arithmetic with complex numbers

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. **5** ✓

b. 7

c. 12

d. 25

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

25 [Foundational] • Target: Complex roots of a quadratic

For which value of k does $x^2 + kx + 16 = 0$ have complex (non-real) roots?

a. $k = 7$ ✓

b. $k = 9$

c. $k = 8$

d. $k = -10$

Complex roots require $k^2 - 64 < 0$, i.e. $-8 < k < 8$. Only $k = 7$ is in that range.

26 [Foundational] • Target: Modulus

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7 + 4i}{2 - i}$?

a. $2 + 3i$ ✓

b. $2 - 3i$

c. $\frac{10 + 15i}{3}$

d. $\frac{18 + 15i}{5}$

To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is $2 + i$: $\frac{7 + 4i}{2 - i} = \frac{(7 + 4i)(2 + i)}{(2 - i)(2 + i)}$ Expand the numerator using the distributive property (FOIL): $(7 + 4i)(2 + i) = 14 + 7i + 8i + 4i^2$ Since $i^2 = -1$: $14 + 15i + 4(-1) = 14 - 4 + 15i = 10 + 15i$ Expand the denominator using the difference of squares: $(2 - i)(2 + i) = 2^2 - i^2 = 4 - (-1) = 5$ Now, divide each term in the num.

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27 [Foundational] • Target: Arithmetic with complex numbers

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.

a. 5

b. **6** ✓

c. 7

d. 8

Let $w = (a + bi) + (3 - 2i) = (a + 3) + (b - 2)i$. We are given that $|w|^2 = 25$, so $(a + 3)^2 + (b - 2)^2 = 25$. Since w lies in the first quadrant, $a + 3 > 0$ and $b - 2 > 0$. We need to find integer or simple fractional solutions for $(a + 3)$ and $(b - 2)$ that sum of squares is 25. The possible pairs of squares are $(9, 16)$ and $(16, 9)$. \\ Case 1: $(a + 3)^2 = 9$ and $(b - 2)^2 = 16$. This gives $a + 3 = \pm 3$ and $b - 2 = \pm 4$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 3$ and $b - 2 = 4$. This leads to $a = 0$ and $b = 6$. How.

28 [Foundational] • Target: Complex roots of a quadratic

Let $z_1 = 2 + 3i$ and $z_2 = 1 - i$. Consider the complex number w such that $w = \frac{z_1^2}{z_2}$. If $w = a + bi$, where a and b are real numbers, find the value of b.

a. $\frac{17}{2}$

b. $-\frac{17}{2}$ ✓

c. $\frac{11}{2}$

d. $-\frac{11}{2}$

First, calculate z_1^2: $z_1^2 = (2 + 3i)^2 = 2^2 + 2(2)(3i) + (3i)^2 = 4 + 12i + 9i^2 = 4 + 12i - 9 = -5 + 12i$. Next, divide z_1^2 by z_2: $w = \frac{-5 + 12i}{1 - i}$. To simplify this division, multiply the numerator and the denominator by the conjugate of the denominator, which is $1 + i$: $w = \frac{-5 + 12i}{1 - i} \times \frac{1 + i}{1 + i} = \frac{(-5 + 12i)(1 + i)}{(1 - i)(1 + i)}$. The numerator is: $(-5)(1) + (-5)(i) + (12i)(1) + (12i)(i) = -5 - 5i + 12i + 12i^2 = -5 + 7i - 12 = -17 + 7i$. The den.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Modulus

Let $z = 3 - 4i$. What is the real part of z?

a. **3** ✓ **b.** -4

c. 5 **d.** 7

A complex number is generally expressed in the form $a + bi$, where a is the real part and b is the imaginary part. In the given complex number $z = 3 - 4i$, the real part is the term that does not have i multiplied by it. Therefore, the real part of z is 3.

30 [Intermediate] • Target: Arithmetic with complex numbers

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

a. **5** ✓ **b.** 7

c. 12 **d.** 25

The magnitude of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the magnitude is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

31 [Intermediate] • Target: Complex roots of a quadratic

Solve $x^2 + 10x + 61 = 0$. Which root has positive imaginary part?

a. **-5+6i** ✓ **b.** -5-6i

c. 5+6i **d.** 5-6i

Discriminant $= 10^2 - 4(61) = -144$. $x = \frac{-10 \pm \sqrt{-144}}{2} = -5 \pm 6i$; the root with positive imaginary part is $-5 + 6i$.

32 [Intermediate] • Target: Modulus

Solve $x^2 + 4x + 8 = 0$. Which root has negative imaginary part?

a. $-2 - 2i$ ✓ **b.** $-2 + 2i$

c. $2 - 2i$ **d.** $-4 - 2i$

Discriminant $= 16 - 32 = -16$. $x = \frac{-4 \pm 4i}{2} = -2 \pm 2i$; the root with negative imaginary part is $-2 - 2i$.

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33 [Intermediate] • Target: Arithmetic with complex numbers

What is the complex conjugate of $-3+7i$?

a. $-3-7i$ ✓

b. $3+7i$

c. $3-7i$

d. $-3+7i$

The conjugate flips only the sign of the imaginary part: $-3+7i \rightarrow -3-7i$.

34 [Intermediate] • Target: Complex roots of a quadratic

Find the complex conjugate of $-10+9i$.

a. **-10-9i** ✓

b. 10+9i

c. 10-9i

d. -10+9i

Flip the sign of the imaginary part only: $-10+9i \rightarrow -10-9i$.

35 [Intermediate] • Target: Modulus

What is the real part of $z=-4+9i$?

a. -4 ✓

b. 9

c. 4

d. -9

The real part is the term without i, which is -4.

36 [Intermediate] • Target: Arithmetic with complex numbers

Consider the function $g(x)=ax^2+bx+c$. The graph of this quadratic function is shown below. The vertex of the parabola is at $(-2,4)$. The parabola passes through the point $(0,-8)$. Find the value of the parameter a.

a. 2

b. **-2** ✓

c. 4

d. -4

The general form of a quadratic function is $g(x)=ax^2+bx+c$. The vertex form of a quadratic function is $g(x)=a(x-h)^2+k$, where (h,k) is the vertex. Given the vertex is $(-2,4)$, we can write the function as $g(x)=a(x-(-2))^2+4$, which simplifies to $g(x)=a(x+2)^2+4$. We are also given that the parabola passes through the point $(0,-8)$. We can substitute these coordinates into the vertex form of the equation to solve for a. $-8=a(0+2)^2+4$ $-8=a(2)^2+4$ $-8=4a+4$ Subtract 4.

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37 [Intermediate] • Target: Complex roots of a quadratic

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.

a. 1

b. **-2 ✓**

c. -1

d. 2

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$ and the x-coordinate of the vertex is given as 1. So, $1 = -\frac{b}{2(1)}$. This simplifies to $1 = -\frac{b}{2}$, which means $b = -2$. Alternatively, we can use the vertex form of a quadratic equation: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$ and $(h, k) = (1, -4)$, so $f(x) = 1(x - 1)^2 - 4$. Expanding this, we get $f(x) = (x^2 - 2x + 1) - 4 = x^2 - 2x - 3$. Comparing this to.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus • Cross-Lesson: 4.01 Matrices & Vectors

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?

a. **3 ✓**

b. -3

c. 12

d. 6

The common difference of an arithmetic sequence is the constant value added to each term to get the next term. Observing the number line, we can see the sequence starts at 0 and progresses to 3, then to 6, and finally to 12. The difference between consecutive terms is $3 - 0 = 3$, $6 - 3 = 3$, and $12 - 6 = 6$. However, the diagram shows markings at 0, 3, 6, and 9. Let's re-examine the number line to ensure accuracy. The marks are at 0, 3, 6, and 9. The difference between consecutive terms is $3 - 0 = 3$.

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39 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 5.01 Introductory Calculus

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.

a. **1** ✓

b. -1

c. 2

d. -2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, the function can be written as $f(x) = a(x - 2)^2 - 5$. We are also given that the y-intercept is $(0, -1)$. This means that when $x = 0$, $f(x) = -1$. Substituting these values into the vertex form: $-1 = a(0 - 2)^2 - 5$. This simplifies to $-1 = a(-2)^2 - 5$, which is $-1 = 4a - 5$. Adding 5 to both sides gives $4 = 4a$. Dividing.

40 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 6.01 Probability

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

a. **14** ✓

b. 17

c. 8

d. 11

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n - 1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4 - 1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

41 [Challenging] • Target: Modulus • Cross-Lesson: 2.01 Sequences & Series

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

a. **17** ✓

b. 12

c. 21

d. 35

The formula for the nth term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. Given $a_1 = 7$ and $d = 5$, we want to find a_3. Substitute $n = 3$ into the formula: $a_3 = 7 + (3 - 1) \times 5$ $a_3 = 7 + (2) \times 5$ $a_3 = 7 + 10$ $a_3 = 17$. Therefore, the 3rd term is 17.

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42 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 3.01 Functions & Graphs

Find the sum of the two roots of $x^2 - 6x + 13 = 0$.

a. 6 ✓

b. 3

c. 13

d. $4i$

By Vieta's formulas the sum of the roots of $x^2 - 6x + 13 = 0$ is $-(-6)/1 = 6$ (the complex parts of the conjugate pair $3 \pm 2i$ cancel).

43 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 4.01 Matrices & Vectors

An arithmetic series has $a_1 = 6$, $d = 2$, and $S_n = 204$. Find n.

a. **12** ✓

b. 13

c. 10

d. 24

Solve $204 = \frac{n}{2}(2(6) + (n-1)(2))$ for n: $n = 12$ (the positive integer root).

44 [Challenging] • Target: Modulus • Cross-Lesson: 5.01 Introductory Calculus

An arithmetic sequence has $a_3 = 0$ and $a_7 = 16$. Find a_1.

a. **-8** ✓

b. -4

c. 8

d. -9

$d = \frac{a_7 - a_3}{7 - 3} = \frac{16 - 0}{4} = 4$. Then $a_1 = a_3 - (3 - 1)d = 0 - (2)(4) = -8$.

45 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 6.01 Probability

Find the common difference of the arithmetic sequence $6,\ 4,\ 2,\ 0,\ \ldots$

a. **-2** ✓

b. 2

c. -1

d. 4

Subtract consecutive terms: $4 - 6 = -2$.

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